

Variational Quantum Eigensolver

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Course Project

Quantum Mechanics for non-physicists (PHYS-344)

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- 1 Introduction to the Algorithm
- 2 VQE Example with Single Qubit Hamiltonian
- 3 Application in Quantum Chemistry
- 4 Caveats

Operator corresponding to the total energy of a quantum system described by Hermitian Matrix.

$$H = H^\dagger \quad (1)$$

Energy of system in state $|\psi\rangle$ is given by the expectation value:

$$E(|\psi\rangle) = \langle\psi|H|\psi\rangle \quad (2)$$

Ground State The lowest energy state $|\psi_0\rangle$ of a quantum system:

$$|\psi_0\rangle = \arg \min_{|\psi\rangle \in \mathcal{H}} E(|\psi\rangle) \quad (3)$$

The Variational Method

Method for approximating the ground state $|\psi_0\rangle$ and lowest energy E_0 of a quantum system¹.

- 1 Choose **ansatz** or **trial state** parameterized by θ :

$$|\psi(\theta)\rangle, \quad \theta = (\theta_1, \theta_2 \dots)$$

- 2 Vary parameters θ to minimize the energy value:

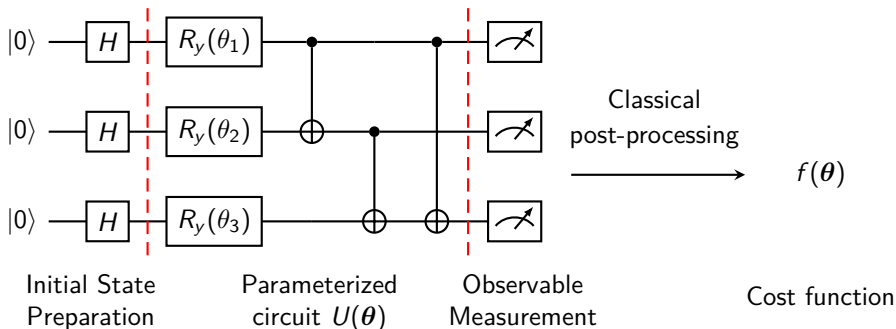
$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle$$
$$\hat{\theta} = \arg \min_{\theta} E(\theta)$$

$E(\hat{\theta})$ is an upper bound for E_0 and $|\psi(\hat{\theta})\rangle \approx |\psi_0\rangle$.

¹Alberto Peruzzo et al. "A variational eigenvalue solver on a photonic quantum processor". In: *Nature communications* 5.1 (2014), p. 4213.

Variational Quantum Circuits

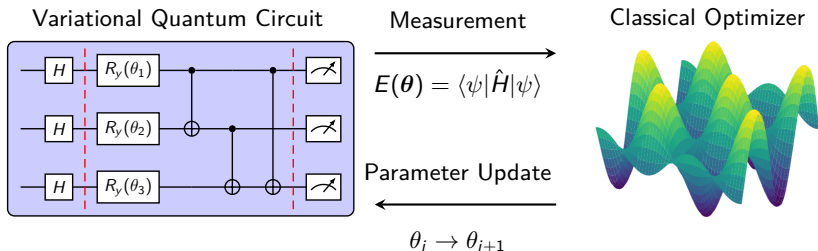
A quantum circuit that depends on parameters $\theta = (\theta_1, \theta_2 \dots)$.



Variational Quantum Eigensolver

Idea: Prepare quantum state with variational quantum circuit $U(\theta)$

$$|\psi(\theta)\rangle = U(\theta) |\psi_{init}\rangle \quad (4)$$



- Use classical-quantum hybrid loop to optimize circuit parameters.
- Similar in structure to classical neural networks.

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VQE Example

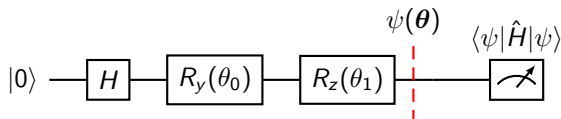
Given a single qubit Hamiltonian,

$$\hat{H} = -\omega_I \hat{I} - \frac{\omega_X}{2} \hat{X} - \frac{\omega_Y}{2} \hat{Y} - \frac{\omega_Z}{2} \hat{Z} \quad (5)$$

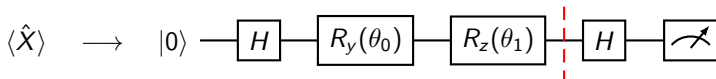
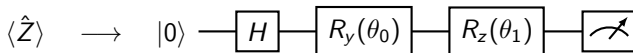
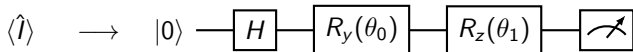
For a given $|\psi\rangle$ we want to measure the expectation value of the Hamiltonian:

$$\langle\psi|H|\psi\rangle = \langle H\rangle = -\omega_I \langle I\rangle - \frac{\omega_X}{2} \langle X\rangle - \frac{\omega_Y}{2} \langle Y\rangle - \frac{\omega_Z}{2} \langle Z\rangle$$

We chose the following Variational Quantum Circuit:

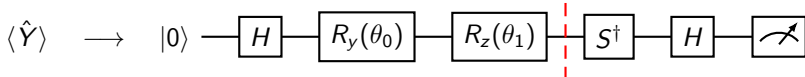


Change of Basis



$$H|0\rangle = |+\rangle,$$

$$H|1\rangle = |-\rangle$$

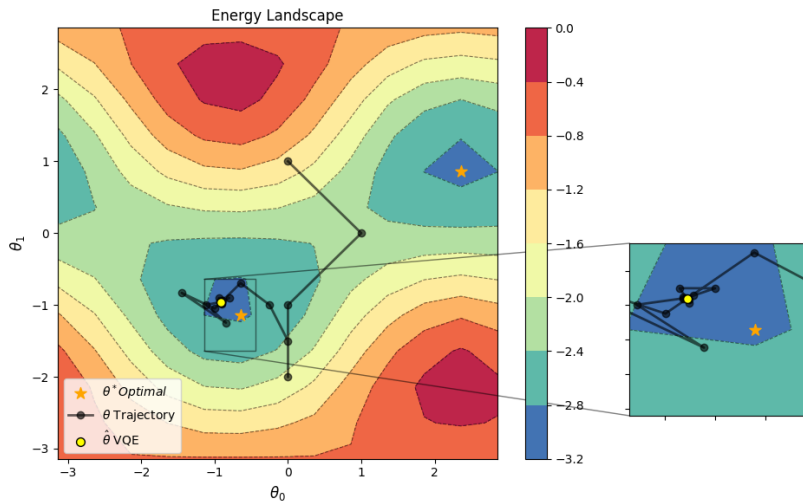


$$SH|0\rangle = |+i\rangle,$$

$$SH|1\rangle = |-i\rangle$$

Note that $\langle \hat{I} \rangle = 1$.

VQE Energy Landscape



- Observe Periodicity in both parameters.
- Classical Optimizer crucial for VQE Success.

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VQE to find GS Energy of H_2

The electronic structure problem

$$\hat{H}_e = -\frac{1}{2} \sum_i \nabla_i^2 - \sum_{i,l} \frac{Z_l}{|\mathbf{r}_i - \mathbf{R}_l|} + \sum_{i \neq j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \quad (6)$$

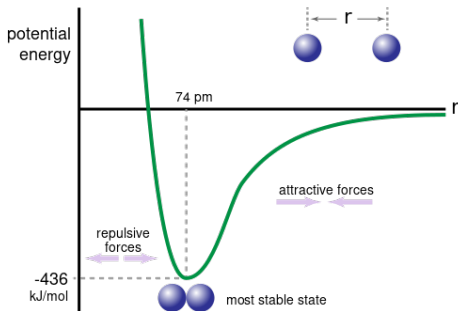
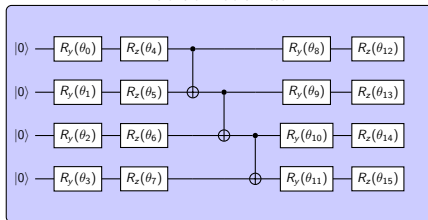


Figure: Potential Energy Curve²

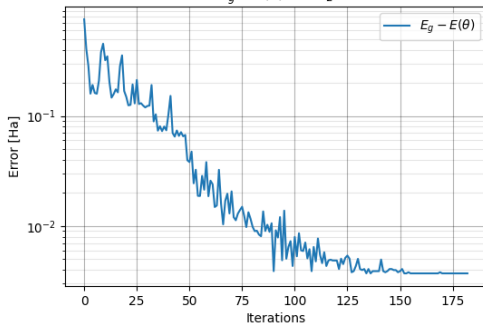
VQE to find GS Energy of H_2

-0.81 IIII
+0.17 IIIZ
-0.23 IIZI
+0.12 IIZZ
+0.17 IZII
+0.17 IZIZ
-0.23 ZIII
+0.17 ZIIZ
+0.05 YYYY
+0.05 XXYY
+0.05 YYXX
+0.05 XXXX
+0.17 IZZI
+0.17 ZIZI
+0.12 ZZII

Hardware Efficient Ansatz



$E_g - E(\theta)$ for H_2



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Caveats

- As variational circuit, VQE has no performance guarantees.
- To estimate the expectation value with a given accuracy ϵ , the number of measurements N must scale as $N \sim 1/\epsilon^2$.

Other contributing factors include

- Limited device connectivity.
- Increasing number of shots and iterations for increasing problem size.

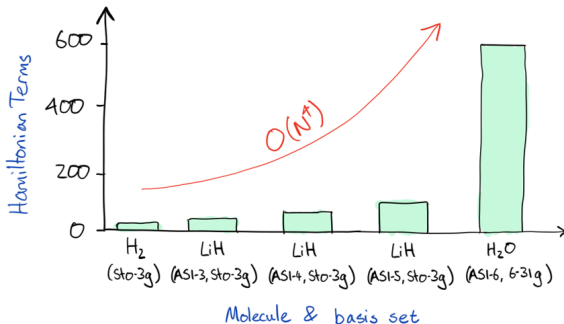


Figure: # of Hamiltonian terms required to represent molecule³

³Josh Izaac. Measurement optimization. https://pennylane.ai/qml/demos/tutorial_measurement_optimize. Date Accessed: 2025-01-04. Jan. 2021.



or
Click to Open in Colab

- [Per+14] Alberto Peruzzo et al. “A variational eigenvalue solver on a photonic quantum processor”. In: *Nature communications* 5.1 (2014), p. 4213.
- [Tex15] University of Texas. *Covalent Bonding*. 2015.
- [Iza21] Josh Izaac. *Measurement optimization*. https://pennyLane.ai/qml/demos/tutorial_measurement_optimize. Date Accessed: 2025-01-04. Jan. 2021.